SCALE-COLLAPSE ALTERNATIVES FOR TYPE II NAVIER–STOKES RESCALINGS

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Abstract. We analyze Type II Navier–Stokes rescalings for which the renormalized scale collapses along the tail. In a scale-translation gauge, the logarithmic scale satisfies $\partial_\tau \log \lambda = a$. If the collapsing rescaling retains a nonzero localized L^2 core, then the negative part of the scale drift has infinite localized accumulation. This gives a scale-drift cost alternative. If instead one seeks a structural PDE description of the same class, then additional compactness, modulation-limit, and omega-limit hypotheses reduce the rescaling to a nonzero stationary solution of a generalized self-similar Navier–Stokes profile equation. In parameter ranges covered by an appropriate Liouville or rigidity theorem, the scale-collapse rescaling is excluded; outside those ranges the remaining question is the corresponding stationary or generalized self-similar Liouville problem.

1. Introduction

Let (u, p) be a suitable weak solution of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0,$$

near a putative Type II singular time. We study a terminal rescaling written in a scale-translation gauge:

$$u(x,t) = \lambda(\tau)^{-1} V\left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau\right), \qquad p(x,t) = \lambda(\tau)^{-2} P\left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau\right) + c(t),$$

where $\lambda(\tau) > 0$. The corresponding renormalized equation is

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0. \tag{1}$$

The convention used throughout is

$$\frac{d}{d\tau} \log \lambda(\tau) = a(\tau). \tag{2}$$

Thus physical scale collapse $\lambda(\tau) \to 0$ is associated with negative logarithmic scale drift.

The preceding reductions isolate compact, radiative, multibubble, and rough-core mechanisms. This paper treats the remaining scale-collapse mechanism in two complementary ways. First, it records the exact finite-or-infinite alternative for the localized negative scale drift. Second, it gives the conditional autonomous-limit reduction for thick scale-collapse windows.

The basic localized mass is

$$M_R(\tau) = \int_{\mathbb{R}^3} |V(y,\tau)|^2 \Phi(\tau, y)\, dy,$$

where $0 \le \Phi \le 1$ is a smooth moving cutoff adapted to the scale-collapse core. The scale-drift density is

$$\mathcal{C}_{\mathrm{sc}}(\tau) = a_-(\tau) M_R(\tau), \qquad a_-(\tau) = \max\{-a(\tau), 0\}.$$

If $\lambda(\tau) \to 0$ and the localized core has a positive L^2 floor, then

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\rm sc}(\tau)\, d\tau = \infty.$$

This gives the localized scale-drift cost alternative.

The same scale-collapse class can also be analyzed through compactness. On selected windows $[\tau_n, \tau_n + T]$, if the translated solutions are locally compact and if

$$a(\tau_n + \cdot) \to a_\infty, \qquad b(\tau_n + \cdot) \to b_\infty$$

in a topology strong enough to pass the coefficient terms in distributions, then a subsequence converges to a nonzero solution of the autonomous equation

$$\partial_s U + (U \cdot \nabla)U + \nabla \Pi = \nu \Delta U + a_\infty(U + y \cdot \nabla U) + b_\infty \cdot \nabla U, \qquad \nabla \cdot U = 0. \tag{3}$$

With an additional stationary omega-limit hypothesis, this yields a nonzero stationary profile (W, Q) satisfying

$$(W \cdot \nabla)W + \nabla Q = \nu \Delta W + a_\infty(W + y \cdot \nabla W) + b_\infty \cdot \nabla W, \qquad \nabla \cdot W = 0. \tag{4}$$

The exclusion is then supplied by the relevant Liouville or rigidity theorem for (4). In the classical backward self-similar normalization this is the role of the theorem of Necas–Ruzicka–Sverak [NRS96]. Other parameter ranges must be stated as separate Liouville hypotheses; forward self-similar Navier–Stokes profiles, for example, are not excluded by a blanket nonexistence statement.

Theorem 1.1 (Main scale-collapse alternative)**.** *Let (V, P, a, b, λ) satisfy (1) and (2) on $[\tau_0, \infty)$. Assume $\lambda(\tau) \to 0$ and assume that the localized core has a positive L^2 floor. Then the localized negative scale-drift cost has infinite total accumulation:*

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

If, in addition, there are thick scale-collapse windows with local compactness, convergent modulation coefficients, and a stationary omega-limit, then the rescaling produces a nonzero stationary solution of (4). Consequently the scale-collapse rescaling is excluded in every parameter regime where the corresponding stationary or generalized self-similar Liouville theorem rules out such nonzero profiles in the admissible class.

2. Scale drift and localized mass

Definition 2.1 (Localized core mass)**.** Let $\Phi(\tau, \cdot) \in C_c^\infty(\mathbb{R}^3)$ be a nonnegative moving cutoff. The associated localized core mass is

$$M_R(\tau) = \int_{\mathbb{R}^3} |V(y, \tau)|^2 \Phi(\tau, y)\, dy.$$

We say that the rescaling has a nonvanishing localized core if there are $m_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$M_R(\tau) \geq m_0 \qquad \text{for a.e. } \tau \geq \tau_1.$$

We say that the localized core is bounded above on a family of windows if $M_R(\tau) \leq M_1$ a.e. on those windows for some $M_1 < \infty$.

Definition 2.2 (Localized scale-drift cost)**.** The localized scale-drift density and its cumulative cost are

$$\mathcal{C}_{\rm sc}(\tau) = a_-(\tau) M_R(\tau), \qquad \mathcal{C}_{\rm sc}[\tau_0, \infty) = \int_{\tau_0}^{\infty} \mathcal{C}_{\rm sc}(\tau)\, d\tau.$$

The density is nonnegative and measures the inward logarithmic scale drift carried by the localized L^2 core.

Lemma 2.3 (Finite-or-infinite alternative)**.** *For every measurable renormalized solution for which a_-M_R is measurable and nonnegative, exactly one of the following alternatives holds:*

$$\int_{\tau_0}^{\infty} a_-(\tau)M_R(\tau)\,d\tau < \infty, \qquad \text{or} \qquad \int_{\tau_0}^{\infty} a_-(\tau)M_R(\tau)\,d\tau = \infty.$$

Proof. The integral is an extended nonnegative number. Hence it is either finite or infinite, and the two cases are mutually exclusive. □

Lemma 2.4 (Finite negative drift with bounded core mass)**.** *Assume $M_R(\tau) \le M_*$ for a.e. $\tau \ge \tau_0$ and*

$$\int_{\tau_0}^{\infty} a_-(\tau)\,d\tau < \infty.$$

Then

$$\int_{\tau_0}^{\infty} a_-(\tau)M_R(\tau)\,d\tau < \infty.$$

Proof. Since $a_- \ge 0$,

$$\int_{\tau_0}^{\infty} a_-(\tau)M_R(\tau)\,d\tau \le M_* \int_{\tau_0}^{\infty} a_-(\tau)\,d\tau < \infty.$$

□

Lemma 2.5 (Logarithmic scale identity)**.** *Under the convention (2), for every $\tau_0 \le \tau_1 < \tau_2 < \infty$,*

$$\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau = \log\lambda(\tau_2) - \log\lambda(\tau_1).$$

Proof. This is the fundamental theorem of calculus applied to $\partial_\tau \log\lambda = a$. □

Theorem 2.6 (Collapse with a nonvanishing core forces infinite scale-drift cost)**.** *Assume*

$$\lambda(\tau) \to 0 \qquad \text{as } \tau \to \infty,$$

and assume that the rescaling has a nonvanishing localized core. Then

$$\int_{\tau_0}^{\infty} a_-(\tau)M_R(\tau)\,d\tau = \infty.$$

Proof. By Theorem 2.5,

$$\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau = \log\lambda(\tau_2) - \log\lambda(\tau_1).$$

Since $\lambda(\tau) \to 0$, the right-hand side tends to $-\infty$ as $\tau_2 \to \infty$. If $\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau < \infty$, then

$$\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau = \int_{\tau_1}^{\tau_2} a_+(\tau)\,d\tau - \int_{\tau_1}^{\tau_2} a_-(\tau)\,d\tau \ge -\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau > -\infty,$$

contradicting the divergence to $-\infty$. Thus $\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty$.

The core floor gives $M_R(\tau) \ge m_0$ for a.e. $\tau \ge \tau_1$. Hence

$$\int_{\tau_0}^{\infty} a_-(\tau)M_R(\tau)\,d\tau \ge m_0 \int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty.$$

□

3. Finite-cost classes and scale-collapse costs

The previous theorem gives an infinite nonnegative quantity. To use this quantity as an exclusion mechanism, the finite-cost class under consideration must be controlled by that quantity. We state this as an analytic hypothesis on the class of scale-collapse rescalinges.

Assumption 3.1 (Finite-cost class controlled by scale drift)**.** For the scale-collapse class under consideration, membership in the finite-cost class implies finiteness of the nonnegative measurable functional

$$\mathcal{C}_{\mathrm{sc}}[\tau_0,\infty) = \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau$$

on the rescaling. Equivalently, a rescaling for which $\mathcal{C}_{\mathrm{sc}}[\tau_0,\infty) = \infty$ does not belong to the finite-cost class.

Theorem 3.2 (Cost exclusion for scale collapse)**.** *Assume that the finite-cost class is controlled by scale drift. If*

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty,$$

then the finite-cost scale-collapse rescaling is excluded.

Proof. The divergent localized scale-drift cost is incompatible with membership in the finite-cost class by Theorem 3.1. □

Definition 3.3 (Absolute scale cost)**.** Let

$$\mathcal{C}_{\mathrm{abs}}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \Phi(\tau,y)\, dy + |a(\tau)| M_R(\tau).$$

Lemma 3.4 (Scale-drift divergence implies absolute cost divergence)**.** *If*

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty,$$

then

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau)\, d\tau = \infty.$$

Proof. Pointwise,

$$\mathcal{C}_{\mathrm{abs}}(\tau) \geq |a(\tau)| M_R(\tau) \geq a_-(\tau) M_R(\tau).$$

Integrating gives the claim. □

Corollary 3.5 (Absolute cost exclusion)**.** *If the finite-cost class is controlled by $\int \mathcal{C}_{\mathrm{abs}}$, then every rescaling satisfying the conclusion of Theorem 2.6 is excluded from the finite-cost class.*

Proof. Apply Theorem 3.4 and then the assumed absolute cost criterion. □

4. Thick drift and autonomous modulation

The infinite weighted drift in Theorem 2.6 does not by itself produce a fixed-length window with a uniform negative average. A bounded nonnegative function may have infinite integral while its unit-window averages tend to zero. The autonomous self-similar reduction therefore requires a separate thickness hypothesis.

Definition 4.1 (Thick negative-drift windows)**.** The rescaling has thick negative-drift windows if there are $T > 0$, $c > 0$, and $\tau_n \to \infty$ such that

$$\frac{1}{T} \int_{\tau_n}^{\tau_n + T} a_-(\tau) M_R(\tau)\, d\tau \geq c \qquad \text{for every } n.$$

Definition 4.2 (Autonomous modulation limit)**.** On the same windows $[\tau_n, \tau_n + T]$, assume there are constants $a_\infty \in \mathbb{R}$ and $b_\infty \in \mathbb{R}^3$ such that

$$a(\tau_n + \cdot) \to a_\infty, \qquad b(\tau_n + \cdot) \to b_\infty$$

in a coefficient topology sufficient to pass the products $a(V + y \cdot \nabla V)$ and $b \cdot \nabla V$ to the limit in distributions. Strong convergence in $L^1(0, T)$ is sufficient.

Lemma 4.3 (Thick drift gives a negative autonomous scale parameter)**.** *Assume thick negative-drift windows, assume $M_R \le M_1$ on these windows, and assume $a(\tau_n + \cdot) \to a_\infty$ in $L^1(0, T)$. Then*

$$a_\infty \le -c/M_1 < 0.$$

Proof. The upper mass bound and thick weighted drift give

$$\frac{1}{T} \int_{\tau_n}^{\tau_n + T} a_-(\tau)\, d\tau \ge c/M_1.$$

After translation and $L^1(0, T)$ convergence of a to the constant a_∞, the left-hand side converges to $(-a_\infty)_+$. Therefore $(-a_\infty)_+ \ge c/M_1$, which gives the claim. □

5. Autonomous limit reduction

Assumption 5.1 (Compactness on selected windows)**.** For the selected windows $[\tau_n, \tau_n + T]$, define

$$V_n(y, s) = V(y, \tau_n + s), \qquad P_n(y, s) = P(y, \tau_n + s), \qquad 0 < s < T.$$

For every bounded ball B_R, the sequence (V_n) is precompact strongly in $L^2(0, T; L^q(B_R))$ for every $q < 6$, weakly compact in $L^2(0, T; H^1(B_R))$, and weak-* compact in $L^\infty(0, T; L^3(\mathbb{R}^3))$. The pressures are normalized modulo functions of time and converge locally in a topology sufficient to pass

$$-\Delta P = \partial_i \partial_j (V_i V_j)$$

to the limit.

Theorem 5.2 (Autonomous reduced limit)**.** *Assume the bounded critical norm*

$$\sup_{\tau \ge \tau_0} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty,$$

assume compactness on selected windows, assume the autonomous modulation limit, and assume that a localized L^2 floor passes to the limit on a fixed compact subcutoff. Then, after passing to a subsequence, (V_n, P_n) converges to a nonzero weak solution (U, Π) of (3) on $\mathbb{R}^3 \times (0, T)$. Moreover

$$U \in L^\infty(0, T; L^3(\mathbb{R}^3)),$$

and the pressure satisfies

$$-\Delta \Pi = \partial_i \partial_j (U_i U_j)$$

modulo functions of s.

Proof. The weak-* $L^\infty L^3$ bound follows from the bounded critical norm. The local $L^2 H^1$ bound and time compactness are included in Theorem 5.1, so Aubin–Lions compactness gives strong local convergence in $L^2(0, T; L^q(B_R))$ for every $q < 6$. This strong local convergence, together with weak $L^2 H^1$ convergence, passes the nonlinear term to $(U \cdot \nabla)U$ in distributions. The pressure convergence passes the Poisson reconstruction to Π.

The autonomous modulation limit passes the coefficient terms to $a_\infty(U + y \cdot \nabla U)$ and $b_\infty \cdot \nabla U$. Hence the limit solves (3). The localized L^2 floor passes to the limit by the assumed convergence on the compact subcutoff, so $U \not\equiv 0$. □

6. Stationary profiles and rigidity

The autonomous limit theorem gives a solution on translated finite windows. It does not by itself produce a stationary profile. That step requires an additional asymptotic compactness or Lyapunov input.

Assumption 6.1 (Stationary omega-limit)**.** The autonomous limit (U, Π) produced by Theorem 5.2 has a nonzero stationary omega-limit (W, Q) satisfying

$$W \in L^3(\mathbb{R}^3),$$

and

$$(W \cdot \nabla)W + \nabla Q = \nu \Delta W + a_\infty (W + y \cdot \nabla W) + b_\infty \cdot \nabla W, \qquad \nabla \cdot W = 0.$$

The solution belongs to the admissibility class required by the rigidity theorem used for the parameter pair (a_∞, b_∞).

Assumption 6.2 (Stationary rigidity for the limiting parameters)**.** For the parameter pair (a_∞, b_∞), every admissible stationary solution (W, Q) of (4) with $W \in L^3(\mathbb{R}^3)$ is either zero or cannot arise as a terminal Type II scale-collapse profile in the class under consideration.

Theorem 6.3 (Conditional self-similar exclusion)**.** *Assume the hypotheses of Theorem 5.2, the stationary omega-limit assumption, and the stationary rigidity assumption for (a_∞, b_∞). Then the scale-collapse rescaling is excluded in that parameter regime.*

Proof. By Theorem 5.2, the rescaling produces a nonzero autonomous limit. By the stationary omega-limit assumption, this yields a nonzero stationary profile (W, Q) in the admissible class for (a_∞, b_∞). The stationary rigidity assumption says that no such nonzero profile can arise from a terminal Type II scale-collapse profile in the class under consideration. This contradicts the assumption that the rescaling belongs to that class. □

Remark 6.4 (Classical and nonclassical parameter regimes)*.* When $a_\infty < 0$ and the parameters can be reduced to the classical backward self-similar normalization, the needed rigidity is modeled on the Necas–Ruzicka–Sverak theorem excluding nontrivial Leray self-similar profiles in the classical admissible class [NRS96]. For other negative values, for drifted equations with $b_\infty \neq 0$, or for the stationary case $a_\infty = 0$, the paper records the corresponding Liouville question as an explicit assumption unless an appropriate theorem is supplied. Forward self-similar solution classes require special care because nontrivial forward self-similar Navier–Stokes solutions may exist in large-data settings.

7. Summary of the alternatives

The scale-collapse class is therefore divided as follows.

(i) If the localized negative scale-drift cost is finite, then the moving localized energy route may absorb the negative scale term together with the remaining finite error terms.
(ii) If the localized negative scale-drift cost is infinite and the finite-cost class is controlled by that cost, then the finite-cost scale-collapse rescaling is excluded.
(iii) If one does not use the cost criterion but has thick windows, compactness, modulation convergence, a stationary omega-limit, and parameter-specific rigidity, then the rescaling is excluded through the generalized self-similar profile equation.
(iv) If the infinite drift is thin or oscillatory, or if compactness, modulation convergence, the stationary omega-limit, or the relevant Liouville theorem is unavailable, the remaining obstruction is the corresponding PDE hypothesis listed above.

References

[NRS96] Jindrich Necas, Michael Ruzicka, and Vladimir Sverak. On leray's self-similar solutions of the navier–stokes equations. *Acta Mathematica*, 176(2):283–294, 1996.